

CUSTOM x86 SOM / preliminary schematic v0.1

Coffee Lake-S + Q370 + four DDR4 SO-DIMMs + AST2500

SOURCE-BASED ELECTRICAL DRAFT – incomplete circuits are listed on each sheet

CPU / Q370 DMI, clocks and sideband

File: 01_cpu_platform.kicad_sch

CPU remaining debug / reserved contacts

File: 08_cpu_reserved_7.kicad_sch

CPU VR controller / source-based starting circuit

File: 15_vrm_controller.kicad_sch

CPU PEG / two x8 carrier links

File: 02_cpu_peg.kicad_sch

Q370 PCIe lane inventory / draft allocation

File: 09_pch_expansion.kicad_sch

Host GbE PHY / I219 reference circuit

File: 16_host_lan.kicad_sch

DDR4 channel A / two SO-DIMMs

File: 03_ddr_a.kicad_sch

Q370 boot flash, LPC and reset interfaces

File: 10_pch_boot.kicad_sch

AST2500 local DDR4 / pin-level interface

File: 17_bmc_ddr.kicad_sch

DDR4 channel B / two SO-DIMMs

File: 04_ddr_b.kicad_sch

Q370 USB2 and PCIe clocks

File: 11_pch_usb_clocks.kicad_sch

DDR4 supply interfaces, termination and socket power

File: 05_memory_power.kicad_sch

Q370 power-domain interface inventory

File: 12_pch_power.kicad_sch

CPU rails and power-pin inventory

File: 06_cpu_power.kicad_sch

AST2500 host links, boot flash and reset

File: 13_bmc_host.kicad_sch

CPU display and audio interface inventory

File: 07_cpu_display.kicad_sch

EPS12V input and unresolved regulator interfaces

File: 14_input_power.kicad_sch

Known omissions: CPU VRM / sequencer / BMC DDR+power+LAN / host PHY / full PDN / connector pinout / remaining PCH GPIO.

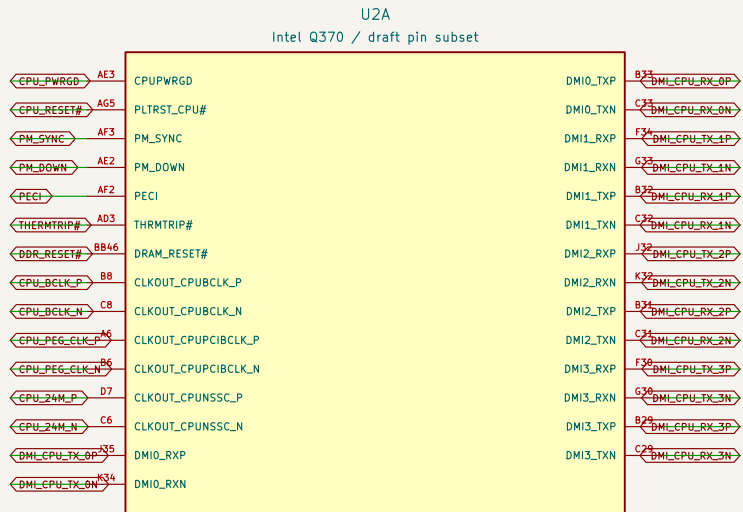
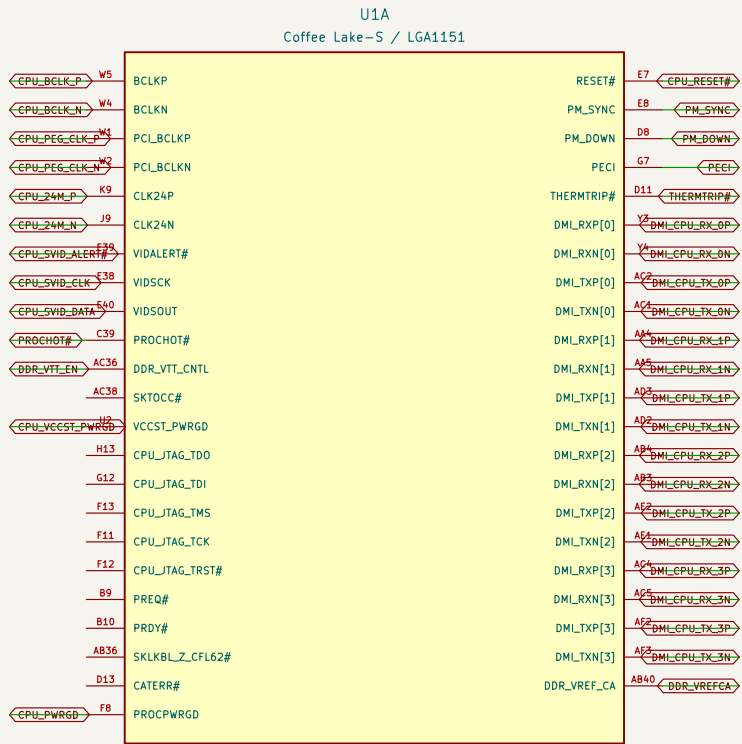
Sheet: /		
File: x86-som-preliminary.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/18

CPU / Q370 DMI, clocks and sideband

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Sources: H310M–52P sheets4–7; Intel 337347 ballout. CPU RX connects PCH TX and vice versa.

DMI is direct as In board reference; package coupling must be checked against the Coffee Lake design guide.



Open: platform pull networks. CPU strap aliases, debug termination and power–good sequencing.

NOT FOR FABRICATION – see open–items.md

Sheet: /CPU / Q370 DMI, clocks and sideband/
File: 01_cpu_platform.kicad_sch

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Size: A2
KiCad E.D.A. 10.0.6

Date:

Rev: 0.1–PRELIMINARY

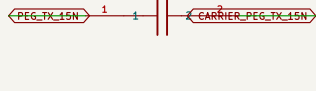
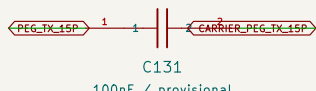
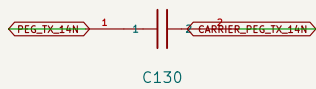
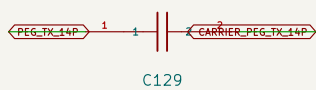
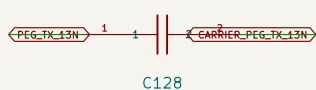
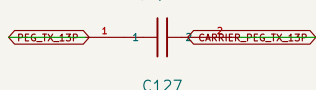
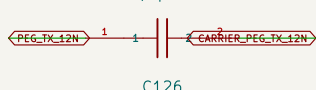
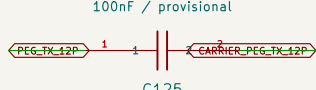
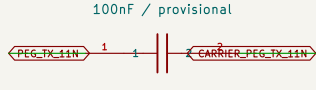
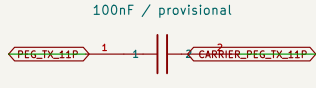
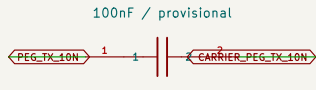
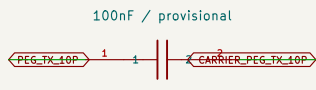
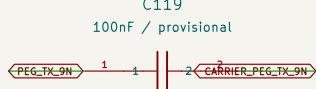
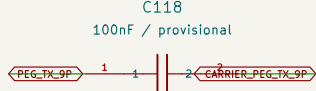
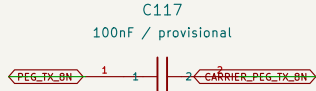
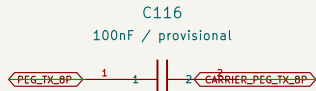
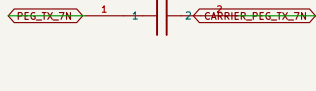
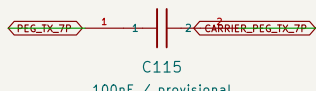
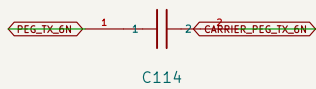
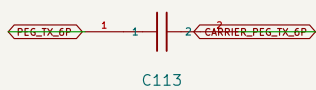
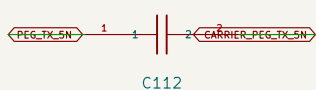
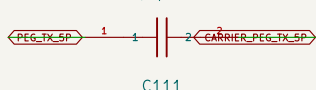
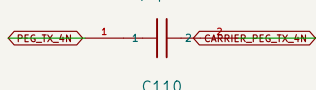
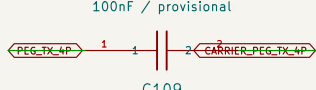
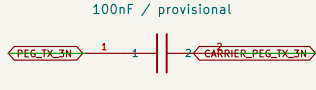
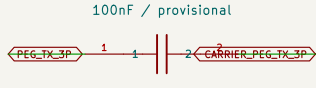
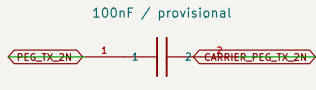
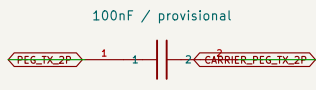
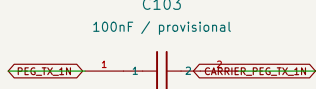
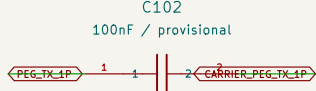
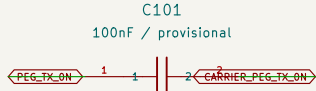
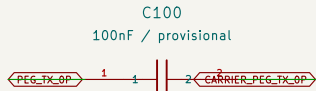
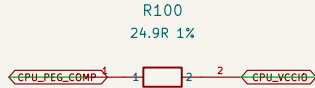
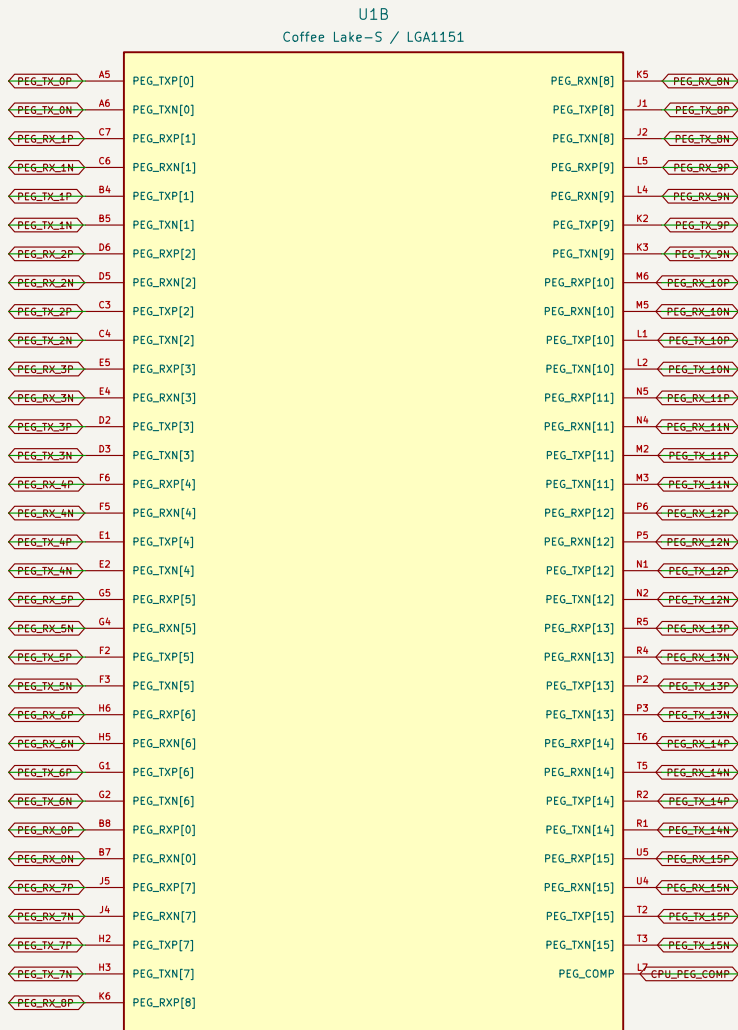
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CPU PEG / two x8 carrier links

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Lanes0–7 → PEG_A x8; lanes8–15 → PEG_B x8. Connector contact numbers remain open.

CPU TX coupling shown below as 100nF candidates; validate placement/value against design guide.



CFG[6:5] = 10 for 2x8. Physical strap aliases still require verification; do not guess pin mapping.

NOT FOR FABRICATION – see open-items.md

Sheet: /CPU PEG / two x8 carrier links/
File: 02_cpu_peg.kicad_sch

Title: CPU PEG / two x8 carrier links

Size: A2
KiCad E.D.A. 10.0.6

Date:

Id: 3/18

Rev: 0.1–PRELIMINARY

DDR4 channel A / two SO-DIMMs

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

CPU pinout: H310M-S2P sheet5, 260–contact connector: GL703GE sheet19, Proposed 2DPC adaptation.

DQ kept logical 1:1, no donor-specific lane swaps. Clock/CS/CKE/ODT split two per socket; layout validation pending.

U1C
Coffee Lake-S / LGA1151

J1A
260-pin DDR4 SO-DIMM

J2A
260-pin DDR4 SO-DIMM

AU33	DDR0_ECC[0]	DDR0_DQSP[8]	AV32
A133	DDR0_ECC[1]	DDR0_DQSN[8]	AU32
AW33	DDR0_ECC[2]	DDR0_DQ[0]	A133
AV31	DDR0_ECC[3]	DDR0_DQ[1]	AW33
AU31	DDR0_ECC[4]	DDR0_DQ[2]	AV31
AV32	DDR0_ECC[5]	DDR0_DQ[3]	AW32
AW31	DDR0_ECC[6]	DDR0_DQ[4]	AU31
A131	DDR0_ECC[7]	DDR0_DQ[5]	AV31
DDR-A-CKP[0] AV18	DDR0_CKP[0]	DDR0_DQ[6]	DDR-A-CKP[0] AV18
DDR-A-CKN[0] AV18	DDR0_CKN[0]	DDR0_DQ[7]	DDR-A-CKN[0] AV18
DDR-A-CKP[1] AV17	DDR0_CKP[1]	DDR0_DQ[8]	DDR-A-CKP[1] AV17
DDR-A-CKN[1] AV17	DDR0_CKN[1]	DDR0_DQ[9]	DDR-A-CKN[1] AV17
DDR-A-CKP[2] AV16	DDR0_CKP[2]	DDR0_DQ[10]	DDR-A-CKP[2] AV16
DDR-A-CKN[2] AV16	DDR0_CKN[2]	DDR0_DQ[11]	DDR-A-CKN[2] AV16
DDR-A-CKP[3] AV16	DDR0_CKP[3]	DDR0_DQ[12]	DDR-A-CKP[3] AV16
DDR-A-CKN[3] AV16	DDR0_CKN[3]	DDR0_DQ[13]	DDR-A-CKN[3] AV16
DDR-A-CKE[0] AV24	DDR0_CKE[0]	DDR0_DQ[14]	DDR-A-CKE[0] AV24
DDR-A-CKE[1] AV24	DDR0_CKE[1]	DDR0_DQ[15]	DDR-A-CKE[1] AV24
DDR-A-CKE[2] AV24	DDR0_CKE[2]	DDR0_DQ[16]	DDR-A-CKE[2] AV24
DDR-A-CKE[3] AV25	DDR0_CKE[3]	DDR0_DQ[17]	DDR-A-CKE[3] AV25
DDR-A-CS# AV12	DDR0_CS#	DDR0_DQ[18]	DDR-A-CS# AV12
DDR-A-CS[1] AV11	DDR0_CS[1]	DDR0_DQ[19]	DDR-A-CS[1] AV11
DDR-A-CS[2] AV13	DDR0_CS[2]	DDR0_DQ[20]	DDR-A-CS[2] AV13
DDR-A-CS[3] AV10	DDR0_CS[3]	DDR0_DQ[21]	DDR-A-CS[3] AV10
DDR-A-ODT[0] AV11	DDR0_ODT[0]	DDR0_DQ[22]	DDR-A-ODT[0] AV11
DDR-A-ODT[1] AV14	DDR0_ODT[1]	DDR0_DQ[23]	DDR-A-ODT[1] AV14
DDR-A-ODT[2] AV12	DDR0_ODT[2]	DDR0_DQ[24]	DDR-A-ODT[2] AV12
DDR-A-ODT[3] AV16	DDR0_ODT[3]	DDR0_DQ[25]	DDR-A-ODT[3] AV16
DDR-A-BA[0] AV13	DDR0_BA[0]	DDR0_DQ[26]	DDR-A-BA[0] AV13
DDR-A-BA[1] AV15	DDR0_BA[1]	DDR0_DQ[27]	DDR-A-BA[1] AV15
DDR-A-BG[0] AV23	DDR0_BG[0]	DDR0_DQ[28]	DDR-A-BG[0] AV23
DDR-A-MA[0] AV13	DDR0_MA[0]	DDR0_DQ[29]	DDR-A-MA[0] AV13
DDR-A-MA[1] AV14	DDR0_MA[1]	DDR0_DQ[30]	DDR-A-MA[1] AV14
DDR-A-MA[2] AV11	DDR0_MA[2]	DDR0_DQ[31]	DDR-A-MA[2] AV11
DDR-A-MA[3] AV15	DDR0_MA[3]	DDR0_DQ[32]	DDR-A-MA[3] AV15
DDR-A-MA[4] AV18	DDR0_MA[4]	DDR0_DQ[33]	DDR-A-MA[4] AV18
DDR-A-MA[5] AV17	DDR0_MA[5]	DDR0_DQ[34]	DDR-A-MA[5] AV17
DDR-A-MA[6] AV19	DDR0_MA[6]	DDR0_DQ[35]	DDR-A-MA[6] AV19
DDR-A-MA[7] AV16	DDR0_MA[7]	DDR0_DQ[36]	DDR-A-MA[7] AV16
DDR-A-MA[8] AV20	DDR0_MA[8]	DDR0_DQ[37]	DDR-A-MA[8] AV20
DDR-A-MA[9] AV20	DDR0_MA[9]	DDR0_DQ[38]	DDR-A-MA[9] AV20
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DDR-A-MA[11] AV20	DDR0_MA[11]	DDR0_DQ[40]	DDR-A-MA[11] AV20
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DDR-A-MA[13] AV22	DDR0_MA[13]	DDR0_DQ[42]	DDR-A-MA[13] AV22
DDR-A-BG[1] AV23	DDR0_BG[1]	DDR0_DQ[43]	DDR-A-BG[1] AV23
DDR-A-ACT# AV24	DDR0_ACT#	DDR0_DQ[44]	DDR-A-ACT# AV24
DDR-A-PAR AV15	DDR0_PAR	DDR0_DQ[45]	DDR-A-PAR AV15
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DDR-A-DQSP[7] AV37	DDR0_DQSP[7]	DDR0_DQ[62]	DDR-A-DQSP[7] AV37

DDR-A-A[0] 144	A0	DQ28	DDR-A-A[0] 144
DDR-A-A[1] 133	A1	DQ29	DDR-A-A[1] 133
DDR-A-A[2] 132	A2	DQ30	DDR-A-A[2] 132
DDR-A-A[3] 131	A3	DQ31	DDR-A-A[3] 131
DDR-A-A[4] 128	A4	DQ32	DDR-A-A[4] 128
DDR-A-A[5] 126	A5	DQ33	DDR-A-A[5] 126
DDR-A-A[6] 127	A6	DQ34	DDR-A-A[6] 127
DDR-A-A[7] 122	A7	DQ35	DDR-A-A[7] 122
DDR-A-A[8] 125	A8	DQ36	DDR-A-A[8] 125
DDR-A-A[9] 121	A9	DQ37	DDR-A-A[9] 121
DDR-A-A[10] 146	A10/AP	DQ38	DDR-A-A[10] 146
DDR-A-A[11] 120	A11	DQ39	DDR-A-A[11] 120
DDR-A-A[12] 119	A12	DQ40	DDR-A-A[12] 119
DDR-A-A[13] 158	A13	DQ41	DDR-A-A[13] 158
DDR-A-A[14] 151	A14/WE#	DQ42	DDR-A-A[14] 151
DDR-A-A[15] 156	A15/CAS#	DQ43	DDR-A-A[15] 156
DDR-A-A[16] 152	A16/RAS#	DQ44	DDR-A-A[16] 152
S2#/C0 162		DQ45	S2#/C0 162
S3#/C1 165		DQ46	S3#/C1 165
DDR-A-ACT# 114	ACT#	DQ47	DDR-A-ACT# 114
DDR-A-PARITY 143	PARITY	DQ48	DDR-A-PARITY 143
DDR-A-ALERT# 116	ALERT#	DQ49	DDR-A-ALERT# 116
DDR-A-EVENT# 154	EVENT#	DQ50	DDR-A-EVENT# 154
DDR-RESET# 108	RESET#	DQ51	DDR-RESET# 108
DDR-A-BM[0] 150	BA0	DQ52	DDR-A-BM[0] 150
DDR-A-BM[1] 145	BA1	DQ53	DDR-A-BM[1] 145
DDR-A-BG[0] 115	BG0	DQ54	DDR-A-BG[0] 115
DDR-A-BG[1] 113	BG1	DQ55	DDR-A-BG[1] 113
DDR-A-CS[0] 149	S0#	DQ56	DDR-A-CS[0] 149
DDR-A-CS[1] 157	S1#	DQ57	DDR-A-CS[1] 157
DDR-A-CKE[0] 249	CKE0	DQ58	DDR-A-CKE[0] 249
DDR-A-CKE[1] 110	CKE1	DQ59	DDR-A-CKE[1] 110
DDR-A-CKP[0] 37	CK0	DQ60	DDR-A-CKP[0] 37
DDR-A-CKN[0] 39	CK0#	DQ61	DDR-A-CKN[0] 39
DDR-A-CKP[1] 38	CK1	DQ62	DDR-A-CKP[1] 38
DDR-A-CKN[1] 40	CK1#	DQ63	DDR-A-CKN[1] 40
DDR-A-ODT[0] 155	ODT0	DQ64	DDR-A-ODT[0] 155
DDR-A-ODT[1] 151	ODT1	DQ65	DDR-A-ODT[1] 151
SMBCLK 253	SCL	DQ66	SMBCLK 253
SMBDATA 254	SDA	DQ67	SMBDATA 254
SPB-WE 256	SA0	DQ68	SPB-WE 256
SPB 260	SA1	DQ69	SPB 260
SPB 166	SA2	DQ70	SPB 166
DDR-A-DQ[0] 8	DQ0	DQ71	DDR-A-DQ[0] 8
DDR-A-DQ[1] 7	DQ1	DQ72	DDR-A-DQ[1] 7
DDR-A-DQ[2] 20	DQ2	DQ73	DDR-A-DQ[2] 20
DDR-A-DQ[3] 21	DQ3	DQ74	DDR-A-DQ[3] 21
DDR-A-DQ[4] 4	DQ4	DQ75	DDR-A-DQ[4] 4
DDR-A-DQ[5] 3	DQ5	DQ76	DDR-A-DQ[5] 3
DDR-A-DQ[6] 16	DQ6	DQ77	DDR-A-DQ[6] 16
DDR-A-DQ[7] 17	DQ7	DQ78	DDR-A-DQ[7] 17
DDR-A-DQ[8] 28	DQ8	DQ79	DDR-A-DQ[8] 28
DDR-A-DQ[9] 29	DQ9	DQ80	DDR-A-DQ[9] 29
DDR-A-DQ[10] 61	DQ10	DQ81	DDR-A-DQ[10] 61
DDR-A-DQ[11] 62	DQ11	DQ82	DDR-A-DQ[11] 62
DDR-A-DQ[12] 24	DQ12	DQ83	DDR-A-DQ[12] 24
DDR-A-DQ[13] 25	DQ13	DQ84	DDR-A-DQ[13] 25
DDR-A-DQ[14] 38	DQ14	DQ85	DDR-A-DQ[14] 38
DDR-A-DQ[15] 37	DQ15	DQ86	DDR-A-DQ[15] 37
DDR-A-DQ[16] 50	DQ16	DQ87	DDR-A-DQ[16] 50
DDR-A-DQ[17] 49	DQ17	DQ88	DDR-A-DQ[17] 49
DDR-A-DQ[18] 62	DQ18	DQ89	DDR-A-DQ[18] 62
DDR-A-DQ[19] 63	DQ19	DQ90	DDR-A-DQ[19] 63
DDR-A-DQ[20] 60	DQ20	DQ91	DDR-A-DQ[20] 60
DDR-A-DQ[21] 65	DQ21	DQ92	DDR-A-DQ[21] 65
DDR-A-DQ[22] 58	DQ22	DQ93	DDR-A-DQ[22] 58
DDR-A-DQ[23] 69	DQ23	DQ94	DDR-A-DQ[23] 69
DDR-A-DQ[24] 70	DQ24	DQ95	DDR-A-DQ[24] 70
DDR-A-DQ[25] 71	DQ25	DQ96	DDR-A-DQ[25] 71
DDR-A-DQ[26] 83	DQ26	DQ97	DDR-A-DQ[26] 83
DDR-A-DQ[27] 84	DQ27	DQ98	DDR-A-DQ[27] 84

DDR-A-A[0] ¹⁴⁴	A0	DQ28	DDR-A-A[0] ⁶⁶
DDR-A-A[1] ¹³³	A1	DQ29	DDR-A-A[1] ⁶⁷
DDR-A-A[2] ¹³²	A2	DQ30	DDR-A-A[2] ⁷⁹
DDR-A-A[3] ¹³¹	A3	DQ31	DDR-A-A[3] ⁸⁰
DDR-A-A[4] ¹²⁸	A4	DQ32	DDR-A-A[4] ¹⁷⁷
DDR-A-A[5] ¹²⁶	A5	DQ33	DDR-A-A[5] ¹⁷³
DDR-A-A[6] ¹²⁷	A6	DQ34	DDR-A-A[6] ¹⁸⁷
DDR-A-A[7] ¹²²	A7	DQ35	DDR-A-A[7] ¹⁸⁹
DDR-A-A[8] ¹²⁵	A8	DQ36	DDR-A-A[8] ¹⁷⁷
DDR-A-A[9] ¹²¹	A9	DQ37	DDR-A-A[9] ¹⁶⁷
DDR-A-A[10] ¹⁴⁶	A10/AP	DQ38	DDR-A-A[10] ¹⁶⁸
DDR-A-A[11] ¹¹¹	A11	DQ39	DDR-A-A[11] ¹⁸²
DDR-A-A[12] ¹¹⁹	A12	DQ40	DDR-A-A[12] ¹⁹⁹
DDR-A-A[13] ¹⁵⁸	A13	DQ41	DDR-A-A[13] ¹⁹²
DDR-A-A[14] ¹⁵¹	A14/WE#	DQ42	DDR-A-A[14] ²⁰⁷
DDR-A-A[15] ¹⁵⁶	A15/CAS#	DQ43	DDR-A-A[15] ²⁰³
DDR-A-A[16] ¹⁵²	A16/RAS#	DQ44	DDR-A-A[16] ¹⁹¹
S2#/C0 ¹⁶²		DQ45	DDR-A-A[16] ¹⁹⁹
S3#/C1 ¹⁶⁵		DQ46	DDR-A-A[16] ²⁰⁷
ACT# ¹¹⁴		DQ47	DDR-A-A[16] ²⁰⁰
PARITY ¹⁴³		DQ48	DDR-A-A[16] ²¹⁶
ALER# ¹¹⁶	ALER#	DQ49	DDR-A-A[16] ²¹⁷
EVENT# ¹⁵⁴	EVENT#	DQ50	DDR-A-A[16] ²²⁸
RESET# ¹⁰⁸	RESET#	DQ51	DDR-A-A[16] ²²⁷
BA0 ¹⁵⁰	BA0	DQ52	DDR-A-A[16] ²¹¹
BA1 ¹⁴⁵	BA1	DQ53	DDR-A-A[16] ²¹¹
BG0 ¹¹⁵	BG0	DQ54	DDR-A-A[16] ²²²
BG1 ¹¹³	BG1	DQ55	DDR-A-A[16] ²²⁵
S0# ¹⁴⁹	S0#	DQ56	DDR-A-A[16] ²³⁷
S1# ¹⁵⁷	S1#	DQ57	DDR-A-A[16] ²³⁶
CKE0 ²⁴⁹	CKE0	DQ58	DDR-A-A[16] ²⁴⁹
CKE1 ¹¹⁰	CKE1	DQ59	DDR-A-A[16] ²⁵⁹
CK0 ³⁷	CK0	DQ60	DDR-A-A[16] ²³⁷
CK0# ³⁹	CK0#	DQ61	DDR-A-A[16] ²⁶¹
CK1 ³⁸	CK1	DQ62	DDR-A-A[16] ²⁶⁴
CK1# ⁴⁰	CK1#	DQ63	DDR-A-A[16] ²⁶⁴
ODT0 ¹⁵⁵	ODT0	DQ64	DDR-A-A[16] ¹²
ODT1 ¹⁵¹	ODT1	DQ65	DDR-A-A[16] ³⁴
SCL ²⁵³	SCL	DQ66	DDR-A-A[16] ⁸
SDA ²⁵⁴	SDA	DQ67	DDR-A-A[16] ⁸
SA0 ²⁵⁶	SA0	DQ68	DDR-A-A[16] ¹⁷⁰
SA1 ²⁶⁰	SA1	DQ69	DDR-A-A[16] ²⁰⁰
SA2 ¹⁶⁶	SA2	DQ70	DDR-A-A[16] ²³⁶
DQ0 ⁸	DQ0	DQ71	DDR-A-A[16] ²⁵⁷
DQ1 ⁷	DQ1	DQ78	DDR-A-A[16] ⁸⁷
DQ2 ²⁰	DQ2	C80	DDR-A-A[16] ⁹²
DQ3 ²¹	DQ3	C81	DDR-A-A[16] ⁹¹
DQ4 ⁶	DQ4	C82	DDR-A-A[16] ¹⁰¹
DQ5 ³	DQ5	C83	DDR-A-A[16] ¹⁰⁵
DQ6 ¹⁶	DQ6	C84	DDR-A-A[16] ⁸⁸
DQ7 ¹⁷	DQ7	C85	DDR-A-A[16] ⁸⁷
DQ8 ²⁸	DQ8	C86	DDR-A-A[16] ¹⁰⁰
DQ9 ²⁹	DQ9	C87	DDR-A-A[16] ¹⁰⁰
DQ10 ⁵¹	DQ10	DW0	DDR-A-A[16] ¹²
DQ11 ⁶²	DQ11	DW1	DDR-A-A[16] ³³
DQ12 ²⁴	DQ12	DW2	DDR-A-A[16] ⁵⁴
DQ13 ²⁵	DQ13	DW3	DDR-A-A[16] ⁷⁵
DQ14 ³⁸	DQ14	DW4	DDR-A-A[16] ¹⁷⁸
DQ15 ³⁷	DQ15	DW5	DDR-A-A[16] ¹⁹⁵
DQ16 ⁶⁰	DQ16	DW6	DDR-A-A[16] ²²⁰
DQ17 ⁶⁹	DQ17	DW7	DDR-A-A[16] ²⁴¹
DQ18 ⁶²	DQ18	DW8	DDR-A-A[16] ⁹⁶
DQ19 ⁵⁹	DQ19	DQ540	DDR-A-A[16] ¹⁸
DQ20 ⁴⁶	DQ20	DQ581	DDR-A-A[16] ⁸
DQ21 ⁶⁵	DQ21	DQ582	DDR-A-A[16] ⁸
DQ22 ⁵⁸	DQ22	DQ583	DDR-A-A[16] ⁷
DQ23 ⁵⁹	DQ23	DQ584	DDR-A-A[16] ¹⁷
DQ24 ³⁹	DQ24	DQ585	DDR-A-A[16] ¹⁹⁰
DQ25 ⁷¹	DQ25	DQ586	DDR-A-A[16] ²⁰
DQ26 ⁸³	DQ26	DQ587	DDR-A-A[16] ²⁰⁶
DQ27 ⁸⁴	DQ27	DQ588	DDR-A-A[16] ⁹⁵

DDR4 channel B / two SO-DIMMs

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

CPU pinout: H310M-S2P sheet5. 260-contact connector: GL703GE sheet19. Proposed 2DPC adaptation.

DQ kept logical 1:1, no donor-specific lane swaps. Clock/CS/CKE/ODT split two per socket; layout validation pending.

U1D
Coffee Lake-S / LGA1151

DDR-B-DQ[0] ^{A034}	DDR1_DQ[0]	DDR1_ECC[4] ^{AP25} ✗
DDR-B-DQ[1] ^{A035}	DDR1_DQ[1]	DDR1_ECC[5] ^{AP25} ✗
DDR-B-DQ[2] ^{A035}	DDR1_DQ[2]	DDR1_ECC[6] ^{AL25} ✗
DDR-B-DQ[3] ^{A035}	DDR1_DQ[3]	DDR1_ECC[7] ^{AL25} ✗
DDR-B-DQ[4] ^{A035}	DDR1_CKP[0]	ANK6 DDR-B-EKFP[0]
DDR-B-DQ[5] ^{A034}	DDR1_CKN[0]	ANK6 DDR-B-EKN[0]
DDR-B-DQ[6] ^{A034}	DDR1_DQ[6]	DDR1_CKP[1] ^{AP22} DDR-B-EKFP[1]
DDR-B-DQ[7] ^{A034}	DDR1_CKN[1]	AP22 DDR-B-EKN[1]
DDR-B-DQ[8] ^{A035}	DDR1_CKP[2]	ANK2 DDR-B-EKFP[2]
DDR-B-DQ[9] ^{AL35}	DDR1_CKN[2]	ANK2 DDR-B-EKN[2]
DDR-B-DQ[10] ^{A032}	DDR1_CKP[3]	AP14 DDR-B-EKFP[3]
DDR-B-DQ[11] ^{AL32}	DDR1_CKN[3]	AP20 DDR-B-EKN[3]
DDR-B-DQ[12] ^{A034}	DDR1_CKE[0]	AV22 DDR-B-EKE[0]
DDR-B-DQ[13] ^{A034}	DDR1_CKE[1]	AV22 DDR-B-EKE[1]
DDR-B-DQ[14] ^{A031}	DDR1_CKE[2]	AKW6 DDR-B-EKE[2]
DDR-B-DQ[15] ^{A031}	DDR1_CKE[3]	ALU2 DDR-B-EKE[3]
DDR-B-DQ[16] ^{AP35}	DDR1_CS#[0]	AP12 DDR-B-CS[0]
DDR-B-DQ[17] ^{A035}	DDR1_CS#1]	AN15 DDR-B-CS[1]
DDR-B-DQ[18] ^{A032}	DDR1_CS#2]	AN12 DDR-B-CS[2]
DDR-B-DQ[19] ^{A032}	DDR1_CS#3]	AN15 DDR-B-CS[3]
DDR-B-DQ[20] ^{A034}	DDR1_ODT[0]	AN14 DDR-B-ODT[0]
DDR-B-DQ[21] ^{AP34}	DDR1_ODT[1]	AL14 DDR-B-ODT[1]
DDR-B-DQ[22] ^{A031}	DDR1_ODT[2]	AP14 DDR-B-ODT[2]
DDR-B-DQ[23] ^{A031}	DDR1_ODT[3]	AL14 DDR-B-ODT[3]
DDR-B-DQ[24] ^{AL29}	DDR1_MA[16]	AK18 DDR-B-MA[16]
DDR-B-DQ[25] ^{A029}	DDR1_MA[14]	AL12 DDR-B-MA[14]
DDR-B-DQ[26] ^{AP22}	DDR1_MA[15]	AP16 DDR-B-MA[15]
DDR-B-DQ[27] ^{A029}	DDR1_BA[0]	AL16 DDR-B-BA[0]
DDR-B-DQ[28] ^{A028}	DDR1_BA[1]	AK18 DDR-B-BA[1]
DDR-B-DQ[29] ^{AL28}	DDR1_BG[0]	AKW8 DDR-B-BG[0]
DDR-B-DQ[30] ^{A028}	DDR1_MA[0]	AL19 DDR-B-MA[0]
DDR-B-DQ[31] ^{AP28}	DDR1_MA[1]	AL22 DDR-B-MA[1]
DDR-B-DQ[32] ^{A028}	DDR1_MA[2]	AKW2 DDR-B-MA[2]
DDR-B-DQ[33] ^{AL22}	DDR1_MA[3]	AKW3 DDR-B-MA[3]
DDR-B-DQ[34] ^{AK13}	DDR1_MA[4]	AP23 DDR-B-MA[4]
DDR-B-DQ[35] ^{AL22}	DDR1_MA[5]	AL23 DDR-B-MA[5]
DDR-B-DQ[36] ^{AK13}	DDR1_MA[6]	AKW6 DDR-B-MA[6]
DDR-B-DQ[37] ^{AL22}	DDR1_MA[7]	AP26 DDR-B-MA[7]
DDR-B-DQ[38] ^{AK12}	DDR1_MA[8]	ALU26 DDR-B-MA[8]
DDR-B-DQ[39] ^{AL22}	DDR1_MA[9]	AKW27 DDR-B-MA[9]
DDR-B-DQ[40] ^{AL10}	DDR1_MA[10]	AP18 DDR-B-MA[10]
DDR-B-DQ[41] ^{AL10}	DDR1_MA[11]	ALU27 DDR-B-MA[11]
DDR-B-DQ[42] ^{AL67}	DDR1_MA[12]	AP22 DDR-B-MA[12]
DDR-B-DQ[43] ^{AL67}	DDR1_MA[13]	AK15 DDR-B-MA[13]
DDR-B-DQ[44] ^{AL69}	DDR1_BG[1]	AV28 DDR-B-BG[1]
DDR-B-DQ[45] ^{AL69}	DDR1_ACT#	ALU28 DDR-B-ACT#
DDR-B-DQ[46] ^{AL66}	DDR1_PAR	AL22 DDR-B-PARITY
DDR-B-DQ[47] ^{AL66}	DDR1_ALERT#	AP22 DDR-B-ALERT#
DDR-B-DQ[48] ^{AL10}	DDR1_DQSN[0]	AP25 DDR-B-DQSN[0]
DDR-B-DQ[49] ^{AL10}	DDR1_DQSN[1]	AP25 DDR-B-DQSN[1]
DDR-B-DQ[50] ^{AK72}	DDR1_DQSN[2]	ANK6 DDR-B-DQSN[2]
DDR-B-DQ[51] ^{AL67}	DDR1_DQSN[3]	ANK6 DDR-B-DQSN[3]
DDR-B-DQ[52] ^{AKW9}	DDR1_DQSN[4]	ANK6 DDR-B-DQSN[4]
DDR-B-DQ[53] ^{AL9}	DDR1_DQSN[5]	AP25 DDR-B-DQSN[5]
DDR-B-DQ[54] ^{AK6}	DDR1_DQSN[6]	ANK6 DDR-B-DQSN[6]
DDR-B-DQ[55] ^{AL6}	DDR1_DQSN[7]	AP25 DDR-B-DQSN[7]
DDR-B-DQ[56] ^{AL6}	DDR1_DQSP[0]	AP25 DDR-B-DQSP[0]
DDR-B-DQ[57] ^{AL67}	DDR1_DQSP[1]	ANK6 DDR-B-DQSP[1]
DDR-B-DQ[58] ^{AL6}	DDR1_DQSP[2]	AP25 DDR-B-DQSP[2]
DDR-B-DQ[59] ^{AL67}	DDR1_DQSP[3]	ANK6 DDR-B-DQSP[3]
DDR-B-DQ[60] ^{AK72}	DDR1_DQ[60]	ANK6 DDR-B-DQSP[4]
DDR-B-DQ[61] ^{AL60}	DDR1_DQ[61]	AP18 DDR-B-DQSP[5]
DDR-B-DQ[62] ^{AL67}	DDR1_DQSP[6]	ANK6 DDR-B-DQSP[6]
DDR-B-DQ[63] ^{AL6}	DDR1_DQSP[7]	AP25 DDR-B-DQSP[7]
AK26 DDR1_ECC[0]	DDR1_ECC[0]	AK25✗
AK26 DDR1_ECC[1]	DDR1_ECC[1]	AK25✗
AK26 DDR1_ECC[2]	DDR1_ECC[2]	AK25✗
AK25 DDR1_ECC[3]	DDR1_ECC[3]	DDR1_VREF-DQ

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DDR-B-A[0] ¹⁴⁴	A0	DQ28	DDR-B-DQ[28] ⁶⁶
DDR-B-A[1] ¹³³	A1	DQ29	DDR-B-DQ[29] ⁶⁷
DDR-B-A[2] ¹³²	A2	DQ30	DDR-B-DQ[30] ⁷⁶
DDR-B-A[3] ¹³¹	A3	DQ31	DDR-B-DQ[31] ⁸⁰
DDR-B-A[4] ¹²⁸	A4	DQ32	DDR-B-DQ[32] ¹⁷²
DDR-B-A[5] ¹²⁶	A5	DQ33	DDR-B-DQ[33] ¹⁷²
DDR-B-A[6] ¹²⁷	A6	DQ34	DDR-B-DQ[34] ¹⁸²
DDR-B-A[7] ¹²⁷	A7	DQ35	DDR-B-DQ[35] ¹⁸²
DDR-B-A[8] ¹²⁵	A8	DQ36	DDR-B-DQ[36] ¹⁷²
DDR-B-A[9] ¹²¹	A9	DQ37	DDR-B-DQ[37] ¹⁶²
DDR-B-A[10] ¹⁴⁶	A10/AP	DQ38	DDR-B-DQ[38] ¹⁸²
DDR-B-A[11] ¹²⁰	A11	DQ39	DDR-B-DQ[39] ¹⁸²
DDR-B-A[12] ¹¹⁹	A12	DQ40	DDR-B-DQ[40] ¹⁷²
DDR-B-A[13] ¹⁵⁸	A13	DQ41	DDR-B-DQ[41] ¹⁵⁹
DDR-B-A[14] ¹⁵¹	A14/WE#	DQ42	DDR-B-DQ[42] ²⁰²
DDR-B-A[15] ¹⁵⁶	A15/CAS#	DQ43	DDR-B-DQ[43] ²⁰²
DDR-B-A[16] ¹⁵²	A16/RAS#	DQ44	DDR-B-DQ[44] ¹⁵⁹
S2#/C0 ¹⁵²	S2#/C0	DQ45	DDR-B-DQ[45] ¹⁵²
S3#/C1 ¹⁶⁵	S3#/C1	DQ46	DDR-B-DQ[46] ²⁰²
DDR-B-ACT# ¹¹⁴	ACT#	DQ47	DDR-B-DQ[47] ²⁰²
DDR-B-PARITY ¹⁴³	PARITY	DQ48	DDR-B-DQ[48] ²¹²
DDR-B-ALERT# ¹¹⁶	ALERT#	DQ49	DDR-B-DQ[49] ²¹²
DDR-B-EVENT# ⁵⁸	EVENT#	DQ50	DDR-B-DQ[50] ²²²
DDR-B-RESET# ¹⁰⁸	RESET#	DQ51	DDR-B-RESET# ¹⁰⁸
DDR-B-BAD# ¹⁵⁰	BAD	DQ52	DDR-B-BAD# ¹⁵⁰
DDR-B-BA[1] ¹⁴⁵	BA1	DQ53	DDR-B-BA[1] ¹⁴⁵
DDR-B-BG[0] ¹¹⁵	BG0	DQ54	DDR-B-BG[0] ²²²
DDR-B-BG[1] ¹¹³	BG1	DQ55	DDR-B-BG[1] ²²²
S0# ¹⁴⁹	S0#	DQ56	DDR-B-CS[25] ¹⁴⁹
DDR-B-CS[1] ^{51#}	S1#	DQ57	DDR-B-CS[27] ¹⁵⁷
CKE0 ¹⁰⁹	CKE0	DQ58	DDR-B-CKE[0] ¹⁰⁹
CKE1 ¹¹⁰	CKE1	DQ59	DDR-B-CKE[0] ²⁵⁰
DDR-B-EKFP[0] ³⁷	CK0	DQ60	DDR-B-EKFP[0] ³⁷²
DDR-B-EKFP[1] ³⁹	CK0#	DQ61	DDR-B-EKFP[1] ²¹²
DDR-B-EKFP[2] ³⁸	CK1	DQ62	DDR-B-EKFP[2] ²⁴²
DDR-B-EKFP[3] ⁴⁰	CK1#	DQ63	DDR-B-EKFP[3] ²⁴²
DDR-B-ODT[0] ¹⁵⁵	ODT0	DQ64	DDR-B-ODT[0] ¹²
DDR-B-ODT[1] ¹⁵¹	ODT1	DQ65	DDR-B-ODT[1] ¹⁵¹
SDCLK ²⁵³	SCL	DQ66	DDR-B-DQSP[1] ²⁵³
SDDATA ²⁵⁴	SDA	DQ67	DDR-B-DQSP[3] ²⁵⁴
SPB-SVCE ²⁵⁶	SAD	DQ68	DDR-B-DQSP[4] ¹⁷²
SPB-SVCE ²⁶⁰	SA1	DQ69	DDR-B-DQSP[5] ²⁰²
SPB-SVCE ¹⁶⁶	SA2	DQ70	DDR-B-DQSP[6] ²²²
Q0 ⁸	Q00	DQ71	DDR-B-DQ[0] ⁸
DDR-B-DQ[1] ⁷	Q1	DQ78	DDR-B-DQ[1] ⁹⁷
Q0 ²⁰	Q02	C80	DDR-B-DQ[2] ⁹² ✗
DDR-B-DQ[3] ²¹	Q03	C81	DDR-B-DQ[3] ⁹¹ ✗
DDR-B-DQ[4] ⁴	Q04	C82	DDR-B-DQ[4] ¹⁰¹ ✗
DDR-B-DQ[5] ³	Q05	C83	DDR-B-DQ[5] ¹⁰⁵ ✗
DDR-B-DQ[6] ¹⁶	Q06	C84	DDR-B-DQ[6] ⁸⁸ ✗
DDR-B-DQ[7] ¹⁷	Q07	C85	DDR-B-DQ[7] ⁸⁷ ✗
DDR-B-DQ[8] ²⁸	Q08	C86	DDR-B-DQ[8] ¹⁰⁰ ✗
DDR-B-DQ[9] ²⁹	Q09	C87	DDR-B-DQ[9] ¹⁰⁴ ✗
DDR-B-DQ[10] ³¹	DQ10	DM0	DDR-B-DQ[10] ¹² ✗
DDR-B-DQ[11] ³²	DQ11	DM1	DDR-B-DQ[11] ³³ ✗
DDR-B-DQ[12] ²⁴	DQ12	DM2	DDR-B-DQ[12] ⁵⁴ ✗
DDR-B-DQ[13] ²⁵	DQ13	DM3	DDR-B-DQ[13] ⁷⁵ ✗
DDR-B-DQ[14] ³⁸	DQ14	DM4	DDR-B-DQ[14] ¹⁷⁸ ✗
DDR-B-DQ[15] ³⁷	DQ15	DM5	DDR-B-DQ[15] ¹⁹² ✗
DDR-B-DQ[16] ⁵⁰	DQ16	DM6	DDR-B-DQ[16] ²²⁰ ✗
DDR-B-DQ[17] ⁴⁹	DQ17	DM7	DDR-B-DQ[17] ²⁴¹ ✗
DDR-B-DQ[18] ⁶²	DQ18	DM8	DDR-B-DQ[18] ⁹⁶ ✗
DDR-B-DQ[19] ⁶³	DQ19	DQ5#0	DDR-B-DQSN[0] ¹⁴
DDR-B-DQ[20] ⁶⁰	DQ20	DQ5#1	DDR-B-DQSN[1] ⁴⁸
DDR-B-DQ[21] ⁶⁵	DQ21	DQ5#2	DDR-B-DQSN[2] ⁶⁵
DDR-B-DQ[22] ⁵⁸	DQ22	DQ5#3	DDR-B-DQSN[3] ²⁸
DDR-B-DQ[23] ⁶⁹	DQ23	DQ5#4	DDR-B-DQSN[4] ¹²²
DDR-B-DQ[24] ⁷⁰	DQ24	DQ5#5	DDR-B-DQSN[5] ¹⁹⁸
DDR-B-DQ[25] ⁷¹	DQ25	DQ5#6	DDR-B-DQSN[6] ²¹
DDR-B-DQ[26] ⁸¹	DQ26	DQ5#7	DDR-B-DQSN[7] ²³
DDR-B-DQ[27] ⁸⁴	DQ27	DQ5#8	DDR-B-DQSN[8] ⁹⁵ ✗

J4A
260-pin DDR4 SO-DIMM

DDR-B-A[0] ¹⁴⁴	A0	DQ28	DDR-B-DQ[28] ⁶⁶
DDR-B-A[1] ¹³³	A1	DQ29	DDR-B-DQ[29] ⁶⁷
DDR-B-A[2] ¹³²	A2	DQ30	DDR-B-DQ[30] ⁷⁶
DDR-B-A[3] ¹³¹	A3	DQ31	DDR-B-DQ[31] ⁸⁰
DDR-B-A[4] ¹²⁸	A4	DQ32	DDR-B-DQ[32] ¹⁷²
DDR-B-A[5] ¹²⁶	A5	DQ33	DDR-B-DQ[33] ¹⁷²
DDR-B-A[6] ¹²⁷	A6	DQ34	DDR-B-DQ[34] ¹⁸²
DDR-B-A[7] ¹²²	A7	DQ35	DDR-B-DQ[35] ¹⁸²
DDR-B-A[8] ¹²⁵	A8	DQ36	DDR-B-DQ[36] ¹⁷²
DDR-B-A[9] ¹²¹	A9	DQ37	DDR-B-DQ[37] ¹⁶²
DDR-B-A[10] ¹⁴⁶	A10/AP	DQ38	DDR-B-DQ[38] ¹⁸²
DDR-B-A[11] ¹²⁰	A11	DQ39	DDR-B-DQ[39] ¹⁸²
DDR-B-A[12] ¹¹⁹	A12	DQ40	DDR-B-DQ[40] ¹⁷²
DDR-B-A[13] ¹⁵⁸	A13	DQ41	DDR-B-DQ[41] ¹⁵⁹
DDR-B-A[14] ¹⁵¹	A14/WE#	DQ42	DDR-B-DQ[42] ²⁰²
DDR-B-A[15] ¹⁵⁶	A15/CAS#	DQ43	DDR-B-DQ[43] ²⁰²
DDR-B-A[16] ¹⁵²	A16/RAS#	DQ44	DDR-B-DQ[44] ¹⁵⁹
S2#/CO ¹⁶²	S2#/CO	DQ45	DDR-B-DQ[45] ¹⁵²
S3#/CL ¹⁶⁵	S3#/CL	DQ46	DDR-B-DQ[46] ²⁰²
DDR-B-ACT# ¹¹⁴	ACT#	DQ47	DDR-B-DQ[47] ²⁰²
DDR-B-PARITY ¹⁴³	PARITY	DQ48	DDR-B-DQ[48] ²¹²
DDR-B-ALERT# ¹¹⁶	ALERT#	DQ49	DDR-B-DQ[49] ²¹²
DDR-B-EVENT# ⁵⁸	EVENT#	DQ50	DDR-B-DQ[50] ²²²
DDR-B-RESET# ¹⁰⁸	RESET#	DQ51	DDR-B-RESET# ¹⁰⁸
DDR-B-BAD# ¹⁵⁰	BAD	DQ52	DDR-B-BAD# ¹⁵⁰
DDR-B-BA[1] ¹⁴⁵	BA1	DQ53	DDR-B-BA[1] ¹⁴⁵
DDR-B-BG[0] ¹¹⁵	BG0	DQ54	DDR-B-BG[0] ²²²
DDR-B-BG[1] ¹¹³	BG1	DQ55	DDR-B-BG[1] ²²²
S0# ¹⁴⁹	S0#	DQ56	DDR-B-CS[25] ¹⁴⁹
DDR-B-CS[1] ^{51#}	S1#	DQ57	DDR-B-CS[27] ¹⁵⁷
CKE0 ¹⁰⁹	CKE0	DQ58	DDR-B-CKE[0] ¹⁰⁹
CKE1 ¹¹⁰	CKE1	DQ59	DDR-B-CKE[0] ²⁵⁰
DDR-B-EKFP[0] ³⁷	CK0	DQ60	DDR-B-EKFP[0] ³⁷²
DDR-B-EKFP[1] ³⁹	CK0#	DQ61	DDR-B-EKFP[1] ²¹²
DDR-B-EKFP[2] ³⁸	CK1	DQ62	DDR-B-EKFP[2] ²⁴²
DDR-B-EKFP[3] ⁴⁰	CK1#	DQ63	DDR-B-EKFP[3] ²⁴²
DDR-B-ODT[0] ¹⁵⁵	ODT0	DQ64	DDR-B-ODT[0] ¹²
DDR-B-ODT[1] ¹⁵¹	ODT1	DQ65	DDR-B-ODT[1] ¹⁵¹
SDCLK ²⁵³	SCL	DQ66	DDR-B-DQSP[1] ²⁵³
SDDATA ²⁵⁴	SDA	DQ67	DDR-B-DQSP[3] ²⁵⁴
SPB-SVCE ²⁵⁶	SAD	DQ68	DDR-B-DQSP[4] ¹⁷²
SPB-SVCE ²⁶⁰	SA1	DQ69	DDR-B-DQSP[5] ²⁰²
SPB-SVCE ¹⁶⁶	SA2	DQ70	DDR-B-DQSP[6] ²²²
Q0 ⁸	Q00	DQ71	DDR-B-DQ[0] ⁸
DDR-B-DQ[1] ⁷	Q1	DQ78	DDR-B-DQ[1] ⁹⁷ ✗
Q0 ²⁰	Q02	C80	DDR-B-DQ[2] ⁹² ✗
DDR-B-DQ[3] ²¹	Q03	C81	DDR-B-DQ[3] ⁹¹ ✗
DDR-B-DQ[4] ⁴	Q04	C82	DDR-B-DQ[4] ¹⁰¹ ✗
DDR-B-DQ[5] ³	Q05	C83	DDR-B-DQ[5] ¹⁰⁵ ✗
DDR-B-DQ[6] ¹⁶	Q06	C84	DDR-B-DQ[6] ⁸⁸ ✗
DDR-B-DQ[7] ¹⁷	Q07	C85	DDR-B-DQ[7] ⁸⁷ ✗
DDR-B-DQ[8] ²⁸	Q08	C86	DDR-B-DQ[8] ¹⁰⁰ ✗
DDR-B-DQ[9] ²⁹	Q09	C87	DDR-B-DQ[9] ¹⁰⁴ ✗
DDR-B-DQ[10] ⁴¹	D010	D00	DDR-B-DQ[10] ¹² ✗
DDR-B-DQ[11] ⁴²	D011	D01	DDR-B-DQ[11] ³¹ ✗
DDR-B-DQ[12] ²⁴	D012	D02	DDR-B-DQ[12] ⁶⁴ ✗
DDR-B-DQ[13] ²⁵	D013	D03	DDR-B-DQ[13] ⁷⁵ ✗
DDR-B-DQ[14] ³⁸	D014	D04	DDR-B-DQ[14] ¹⁷⁸ ✗
DDR-B-DQ[15] ³⁷	D015	D05	DDR-B-DQ[15] ¹⁹² ✗
DDR-B-DQ[16] ²⁰	D016	D06	DDR-B-DQ[16] ²²⁰ ✗
DDR-B-DQ[17] ⁴⁹	D017	D07	DDR-B-DQ[17] ²⁴¹ ✗
DDR-B-DQ[18] ⁶²	D018	D08	DDR-B-DQ[18] ⁹⁶ ✗
DDR-B-DQ[19] ⁶³	D019	D05#0	DDR-B-DQSP[0] ¹⁶ ✗
DDR-B-DQ[20] ⁴⁰	D020	D05#1	DDR-B-DQSP[1] ¹⁶ ✗
DDR-B-DQ[21] ⁴⁵	D021	D05#2	DDR-B-DQSP[2] ¹⁶ ✗
DDR-B-DQ[22] ⁵⁸	D022	D05#3	DDR-B-DQSP[3] ¹⁶ ✗
DDR-B-DQ[23] ⁵⁹	D023	D05#4	DDR-B-DQSP[4] ¹⁷² ✗
DDR-B-DQ[24] ⁷⁰	D024	D05#5	DDR-B-DQSP[5] ¹⁷² ✗
DDR-B-DQ[25] ⁷¹	D025	D05#6	DDR-B-DQSP[6] ¹⁷² ✗
DDR-B-DQ[26] ⁸³	D026	D05#7	DDR-B-DQSP[7] ²⁷² ✗
DDR-B-DQ[27] ⁸⁴	D027	D05#8	DDR-B-DQSP[8] ⁹⁵ ✗

DDR4 supply interfaces, termination and socket power

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

DDR_1V2 and DDR_VPP_2V5 regulators remain to size for four modules. VTT circuit follows TP551200.

Shown capacitors are preliminary, not a complete PDN. Memory VREF_DQ handling still needs CPU guidance.

J1B
260-pin DDR4 SO-DIMM



J2B
260-pin DDR4 SO-DIMM



J3B
260-pin DDR4 SO-DIMM



J4B
260-pin DDR4 SO-DIMM



U10
TP551200DRC



C200
10uF



C201
10uF



C202
10uF



C203
100nF



C204
1uF



C205
10uF



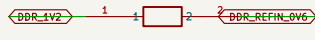
R201
2.2k / provisional



R202
2.2k / provisional



R203
10k 0.1%



R204
10k 0.1%



C206
100nF



SPD_3V3 source/sleep-state policy is open. Shared PCH/BMC SMBus requires ownership and voltage review.

NOT FOR FABRICATION – see open-items.md

Sheet: /DDR4 supply interfaces, termination and socket power/
File: 05_memory_power.kicad_sch

Title: DDR4 supply interfaces, termination and socket power

Size: A2
KiCad E.D.A. 10.0.6

Date:

Id: 6/18

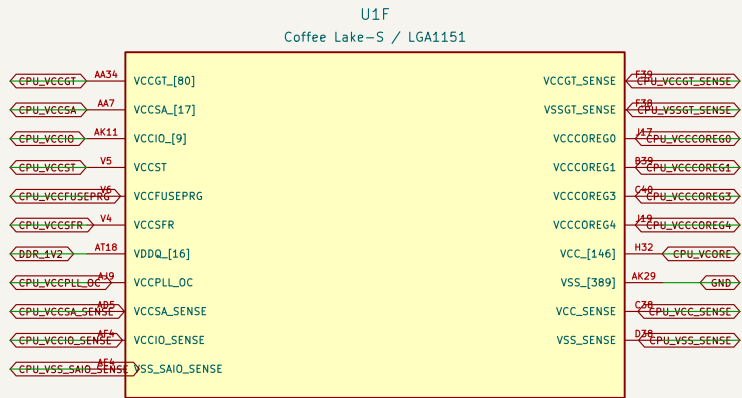
Rev: 0.1 – PRELIMINARY

CPU rails and power-pin inventory

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

All 1151 socket contacts are present across CPU units. Supply pins stack by rail; exact pins are in pin-audit.json.

No CPU VRM is claimed complete. ISL95866 donor sheets25–27 are retained for the next circuit transcription.



Required: CPU_VCORE, VCCGT, VCCSA, VCCIO, VCCST/PLL and DDR_1V2.

Open: VR controller order code, 4+2 vs other phase count, load line, current limit, inductors, MOSFETs, compensation.

Do not connect every rail with similar voltage together: sequencing and isolation are part of the implementation.

NOT FOR FABRICATION – see open-items.md

Sheet: /CPU rails and power-pin inventory/
File: 06_cpu_power.kicad_sch

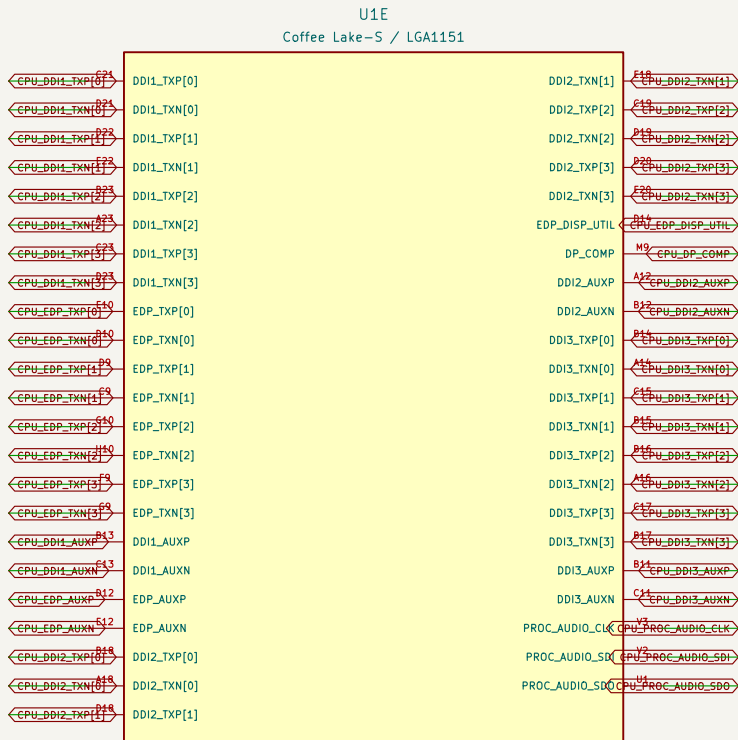
Title: CPU rails and power-pin inventory

Size: A2	Date:	Rev: 0.1–PRELIMINARY
KiCad E.D.A. 10.0.6		Id: 7/18

CPU display and audio interface inventory

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Donor H310M–S2P sheet4. Display port selection and carrier connector allocation are not frozen.



NOT FOR FABRICATION – see open-items.md

Sheet: /CPU display and audio interface inventory/
File: 07_cpu_display.kicad_sch

Title: CPU display and audio interface inventory

Size: A2
KiCad E.D.A. 10.0.6

Date:

Rev: 0.1 – PRELIMINARY

Id: 8/18

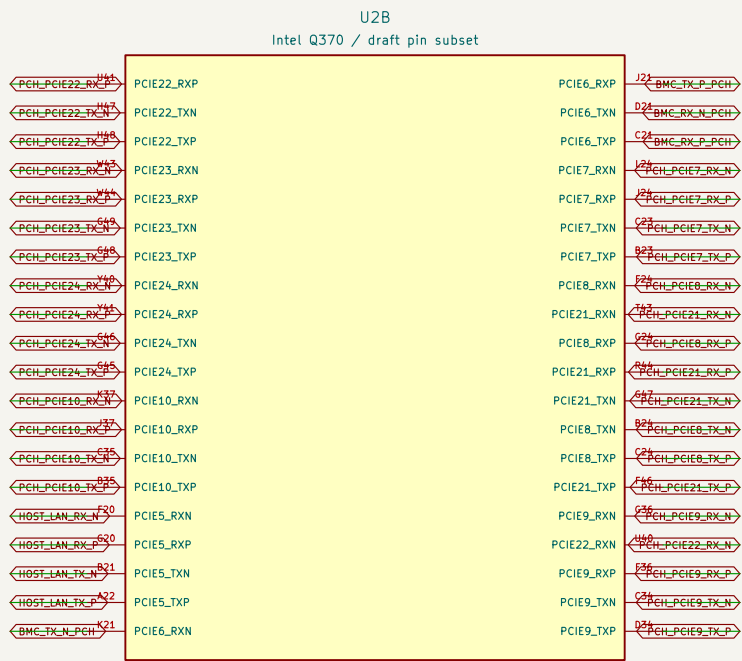


Q370 PCIe lane inventory / draft allocation

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Proposed only: PCIe lane5 (HSIO10) shared with GbE; lane6 AST2500; 1–4,7–12 ten native x1; 13–16,17–20,21–24 three x4.

Numbered allocation must be checked against HSIO mux figure and straps. Carrier switches and connector pins not yet drawn.



No SATA allocation in this PCIe–max draft. Adding SATA requires a new lane budget.

PCH PCIe TX AC coupling remains to add for each external lane.

I219 link must use the selected GbE mux lane; lane5 / HSIO10 selected per Intel Fig3–1; strap verification pending.

NOT FOR FABRICATION – see open-items.md

Sheet: /Q370 PCIe lane inventory / draft allocation/
File: 09_pch_expansion.kicad_sch

Title: Q370 PCIe lane inventory / draft allocation

Size: A2

Date:

KiCad E.D.A. 10.0.6

Rev: 0.1–PRELIMINARY

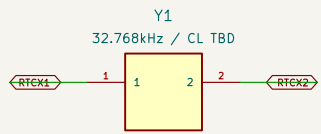
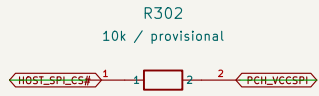
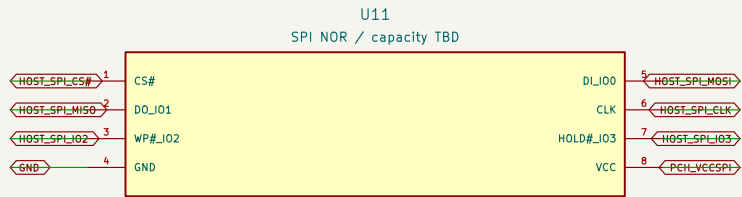
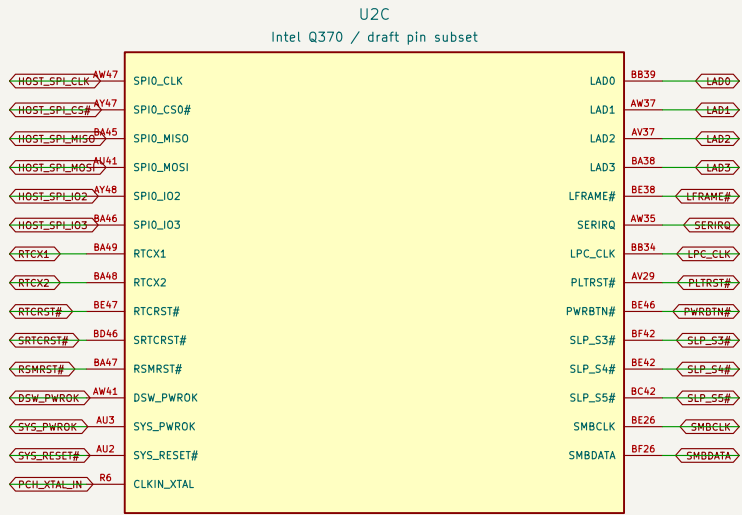
Id: 10/18

Q370 boot flash, LPC and reset interfaces

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Intel 337347 ballout, H310M–S2P sheets10–12 as topology reference only. GPIO aliases require an independent check.

Host flash 3.3V pollyc proposed. BMC BIOS recovery mux is not implemented; flash presently connects to PCH only.



24MHz crystal/oscillator circuit, RTC battery/reset network and reset supervisor remain open.

PWRBTN#, SLP_S3/4/5#, DSW_PWROK, RSMRST# and SYS_PWROK are interfaces, not a completed sequencer.

NOT FOR FABRICATION – see open-items.md

Sheet: /Q370 boot flash, LPC and reset interfaces/
File: 10_pch_boot.kicad_sch

Title: Q370 boot flash, LPC and reset interfaces

Size: A2

Date:

KiCad E.D.A. 10.0.6

Rev: 0.1–PRELIMINARY

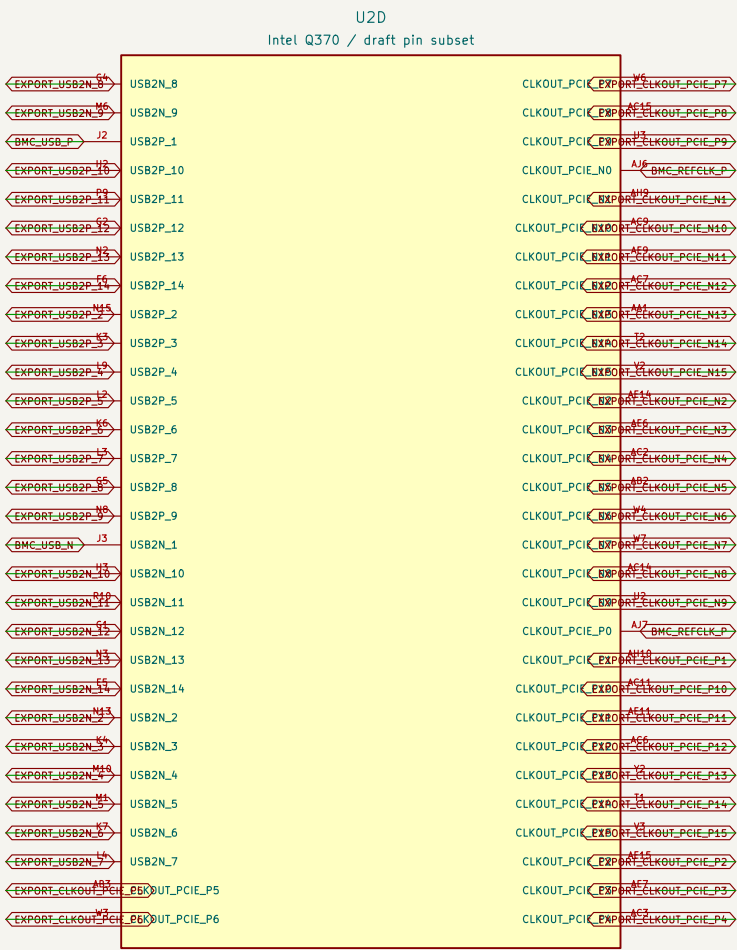
Id: 11/18

Q370 USB2 and PCIe clocks

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

USB2 port1 goes to AST2500 USB–A virtual hub. Other USB2 ports are carrier interface candidates.

Clock output0 reserved for AST2500; remaining reference clocks require CLKREQ# ownership and fanout budget.



NOT FOR FABRICATION – see open-items.md

Sheet: /Q370 USB2 and PCIe clocks/
File: 11_pch_usb_clocks.kicad_sch

Title: Q370 USB2 and PCIe clocks

Size: A2

Date:

KiCad E.D.A. 10.0.6

Rev: 0.1 – PRELIMINARY

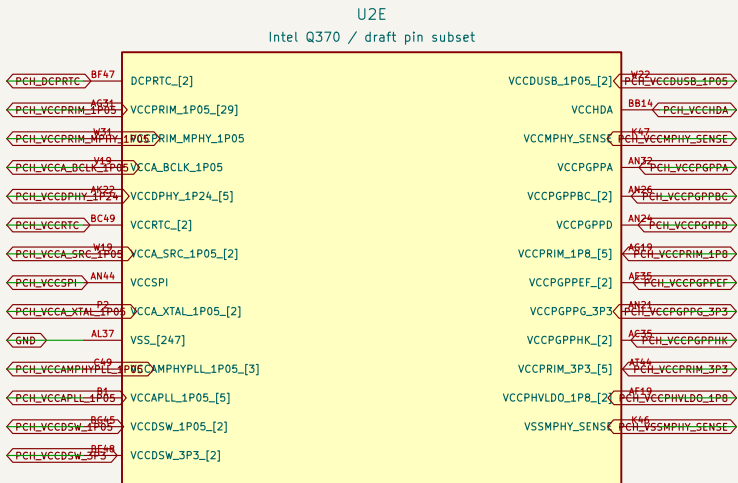
Id: 12/18

Q370 power-domain interface inventory

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Intel 337347 Table8–1 and ballout. Identical rail names stack all extracted power balls; pin audit lists them.

Internal outputs DCPRTC, VCCDPHY_1P24, VCCDSW_1P05 must not be treated as external supply inputs.



Primary/suspend and deep-sleep domains need separate enable-state review. Regulators/filter networks not implemented.

VCCPHVLD0_1P8 wiring depends on internal/external 1.8V mode. Keep separate until mode is selected.

NOT FOR FABRICATION – see open-items.md

Sheet: /Q370 power-domain interface inventory/
File: 12_pch_power.kicad_sch

Title: Q370 power-domain interface inventory

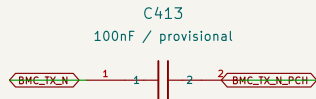
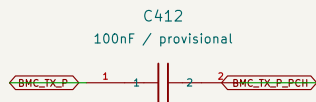
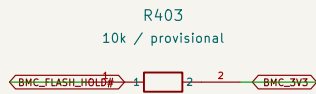
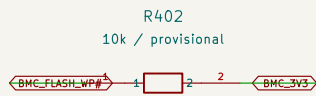
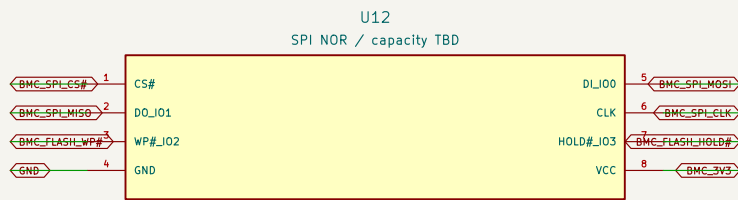
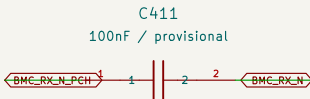
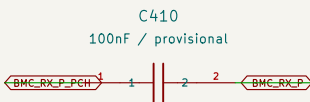
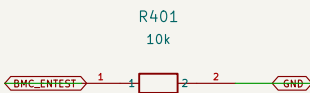
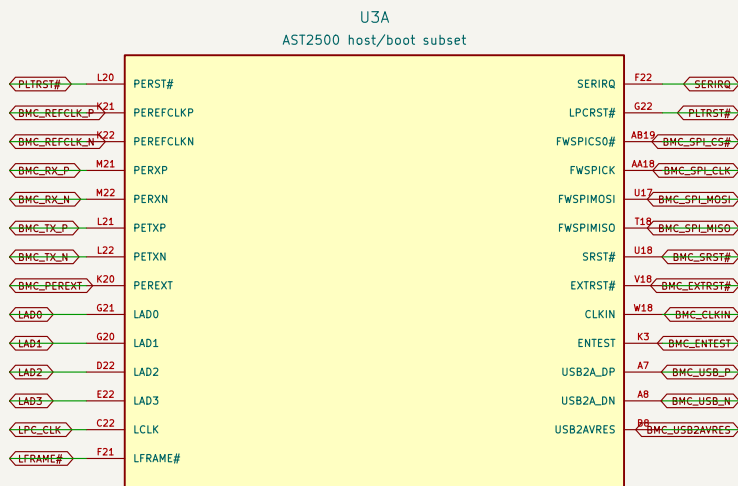
Size: A2	Date:	Rev: 0.1 – PRELIMINARY
KiCad E.D.A. 10.0.6		Id: 13/18

AST2500 host links, boot flash and reset

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

AST2500 A2 v1.6 pp52–53,58,68–69: X115CH–F block diagram corroborates PCIe + LPC + USB integration.

This is the host–interface subset. BMC power, DDR4, GPIO straps, management PHY and firmware are not complete.



CLKIN needs a 24MHz oscillator; SRST# must remain asserted >=1ms after power is stable.

Flash pins are also boot straps; verify strap resistor requirements before adopting these pull–ups.

NOT FOR FABRICATION – see open–items.md

Sheet: /AST2500 host links, boot flash and reset/
File: 13_bmc_host.kicad_sch

Title: AST2500 host links, boot flash and reset

Size: A2
KiCad E.D.A. 10.0.6

Date:

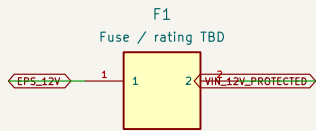
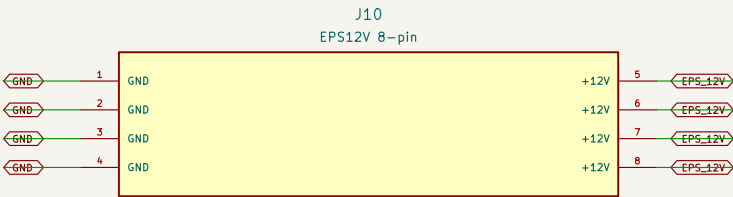
Rev: 0.1–PRELIMINARY
Id: 14/18

EPS12V input and unresolved regulator interfaces

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

8-pin EPS input retained. External 12V must remain available for BMC standby if deriving all standby rails locally.

Protection and rail converters are not sized in this draft. No eFuse/MOSFET ratings are implied.



Next circuit transcription: donor ISL95866 sheets25–27; DDR regulator sheet29; VPP sheet30.

Required independently checked rails: AUX_3V3, BMC_3V3/core/DDR, PCH domains, CPU domains, DDR_1V2/VPP.

No fabrication footprint, regulator compensation, current budget or sequencing approval in v0.1.

NOT FOR FABRICATION – see open-items.md

Sheet: /EPS12V input and unresolved regulator interfaces/
File: 14_input_power.kicad_sch

Title: EPS12V input and unresolved regulator interfaces

Size: A2

Date:

KiCad E.D.A. 10.0.6

Rev: 0.1–PRELIMINARY

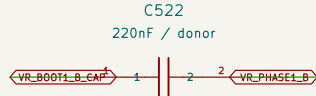
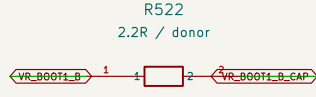
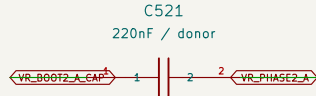
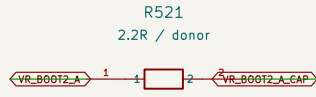
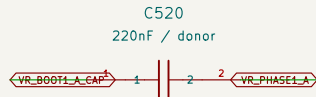
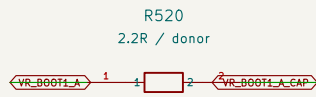
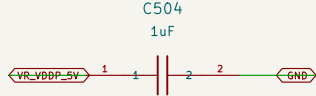
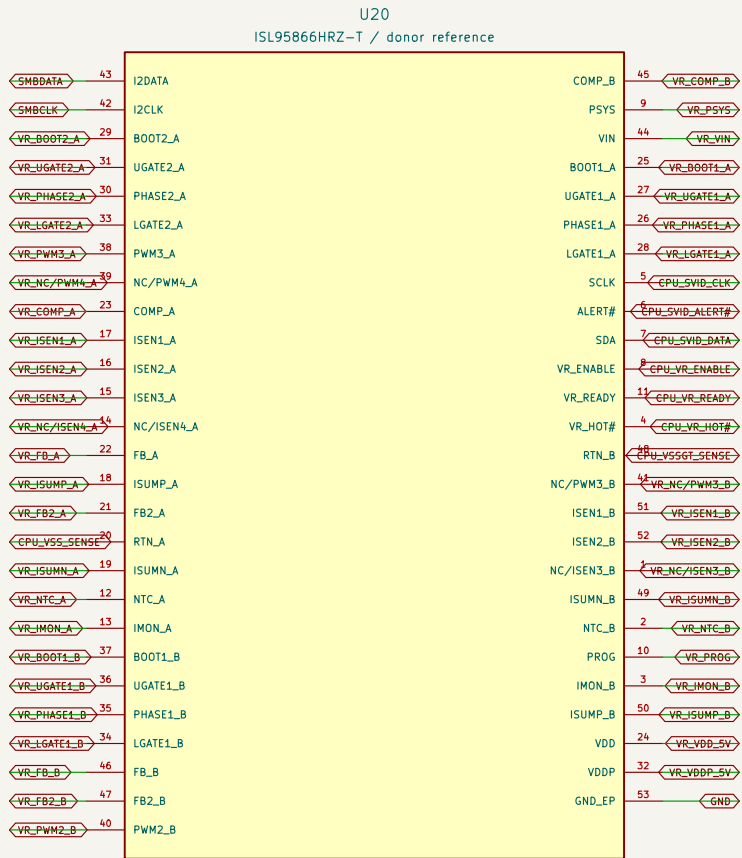
Id: 15/18

CPU VR controller / source-based starting circuit

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Transcribed pinout and bias/bootstrap networks: H310M–S2P 2.0 sheet25. This does not constitute a complete VRM.

Donor contains ISL95856/ISL95866 population alternatives. Phase mapping, compensation and current sense remain open.



UGATE/LGATE/PHASE are unconnected power-stage interfaces. Feedback/current-sense must be completed before power-on.

CPU_VCORE and CPU_VCCGT are not generated in this draft. Donor sheets26–27 retain exact MOSFET/driver circuits.

CPU SVID source pull-up/series networks and PSYS/IMON/NTC programming remain required.

NOT FOR FABRICATION – see open-items.md

Sheet: /CPU VR controller / source-based starting circuit/
File: 15_vrm_controller.kicad_sch

Title: CPU VR controller / source-based starting circuit

Size: A2

Date:

Rev: 0.1–PRELIMINARY

KiCad E.D.A. 10.0.6

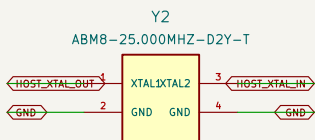
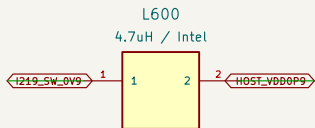
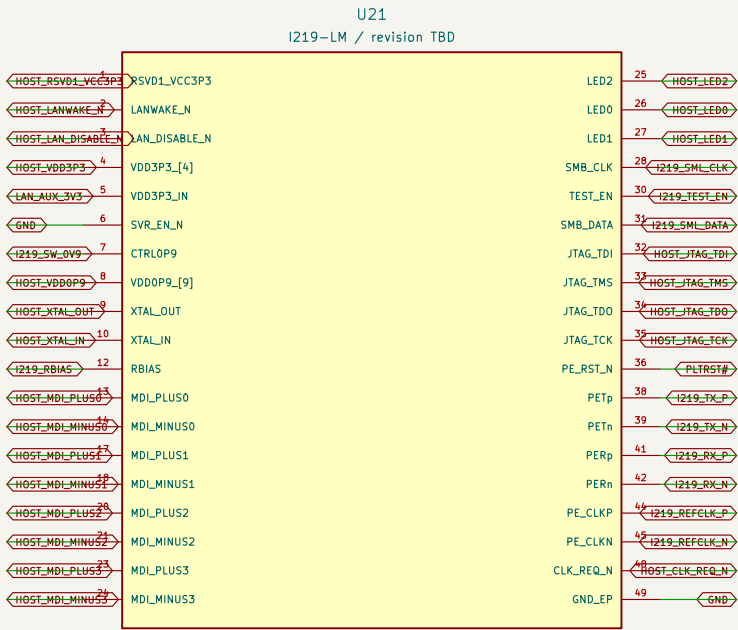
Id: 16/18

Host GbE PHY / I219 reference circuit

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

Intel I219 full datasheet pin tables. Exact I219–LM generation and Q370 compatibility must be selected before BOM freeze.

Uses PCH GbE on HSI010 / PCIe lane5. This lnk consumes the shared lane; It is not an additional expansion root port.



Open: LANPHYPC / LAN_DISABLE logic, SMLink wiring, source clock+CLKREQ, internal regulator caps and strap pull–ups.

MDI pairs go to carrier magnetics/RJ45; no magnetics or termination network is included yet.

NOT FOR FABRICATION – see open–items.md

Sheet: /Host GbE PHY / I219 reference circuit/
File: 16_host_lan.kicad_sch

Title: Host GbE PHY / I219 reference circuit

Size: A2

Date:

KiCad E.D.A. 10.0.6

Rev: 0.1–PRELIMINARY

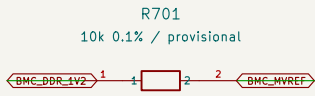
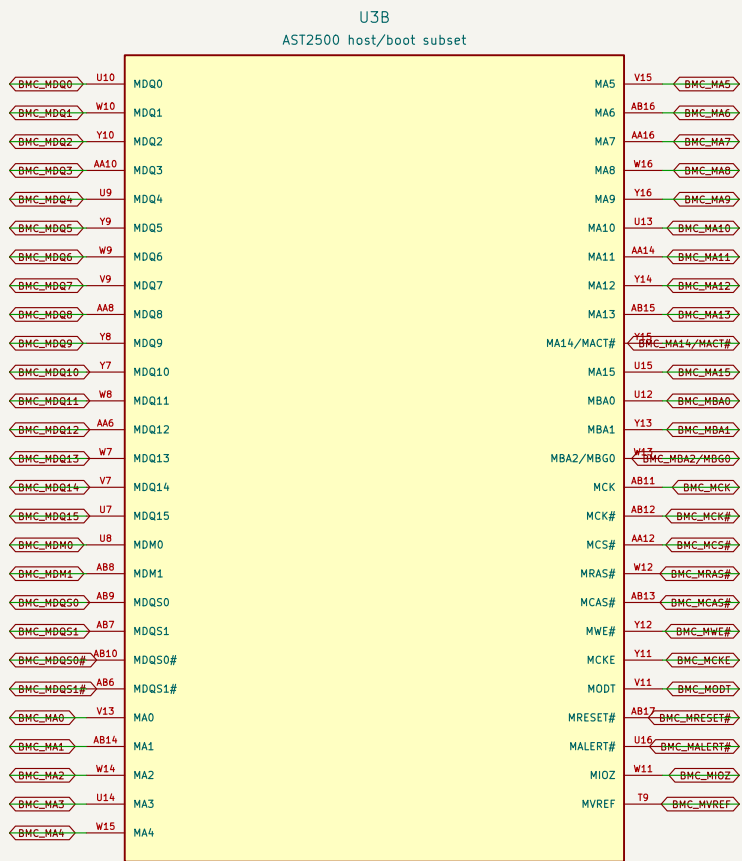
Id: 17/18

AST2500 local DDR4 / pin-level interface

PRELIMINARY v0.1 – electrically incomplete; not for fabrication

AST2500 DDR4 x16 interface from full datasheet pp51–52. Exact DRAM part and footprint remain unselected.

MA14/MACT#, MBA aliases and required memory geometry must be reconciled with chosen DDR4 before connecting its balls.



This sheet is an unfinished memory circuit: 53 BMC interface pins are available, but no RAM component is represented.

BMC core/power domains, RGMII management PHY, reset supervisor and complete strap matrix remain required.

NOT FOR FABRICATION – see open-items.md

Sheet: /AST2500 local DDR4 / pin-level interface/
File: 17_bmc_ddr.kicad_sch

Title: AST2500 local DDR4 / pin-level interface

Size: A2
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Date:

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